Monitoring Kubernetes with Dynatrace Operator

Dynatrace Professional Services Team
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Prepared for

Requirement

- Kubernetes version : 1.21+
- Dynatrace Operator version : 0.4.0+
- Ingress and Egress connection from k8s Cluster to DT Managed/SaaS
- Admin privilege
- Check support lifecycle k8s:

Kubernetes upstream version	OpenShift version	Minimum OneAgent version	Minimum ActiveGate version	Minimum Dynatrace Operator version	Recommended Dynatrace Operator version	End of support
1.27		1.269	1.269	v0.11.2	v0.12.0+	
1.26	4.13	1.259	1.257	v0.11.2	v0.12.0+	
1.25	4.12	1.249	1.251	v0.8.x	v0.12.0+	
1.24	4.11	1.241	1.243	v0.7.x	v0.12.0+	
1.23	4.10	1.233	1.233	v0.4.x	v0.12.0+	
1.22	4.9	1.227	1.223	v0.3.x	v0.12.0+	
1.21	4.8	1.217	1.215	v0.3.x	v0.12.0+	
1.20	4.7	1.207	1.211	v0.3.x	v0.6.0	Aug 1, 2023
1.19	4.6	1.199	1.205	v0.3.x	v0.6.0	Aug 1, 2023

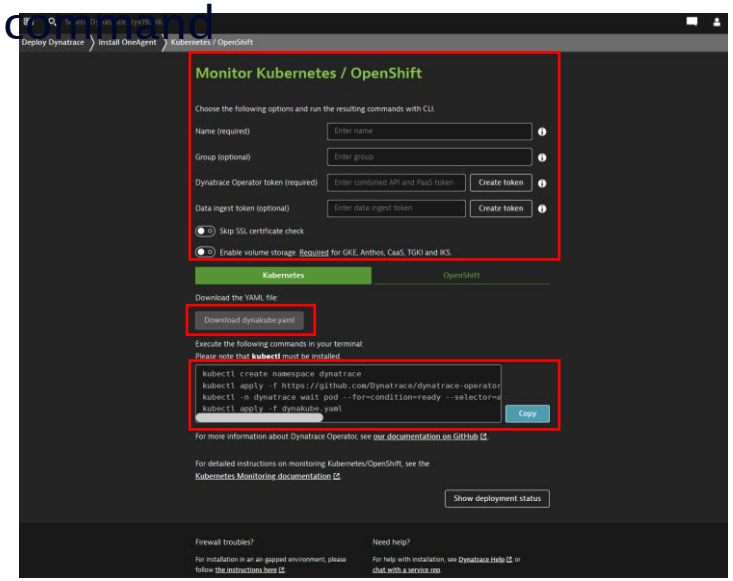
More Reference

Monitor Kubernetes

with Dynatrace Operator

Step 1

Simply go to Deploy Dynatrace and get the dynakube.yaml file from Dynatrace UI and copy below command



Make sure to have kubeadmin access and do some checking regarding version, nodes or etc before apply Dynatrace operator

```
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl version
WARNING: This version information is deprecated and will be replaced with the output from kubectl version --short.  Use --output=yaml|json to get the full version.
Client Version: version.Info{Major:"1", Minor:"25", GitVersion:"v1.25.0", GitCommit:"a866cbe2e5bbaa01cfd5e969aa3e033f3282a8a2", GitTreeState:"clean", BuildDate:"2022-08-23T17:44:59Z", GoVersion:"go1.19", Compiler:"gc", Platform:"linux/amd64"}
Kustomize Version: v4.5.7
Server Version: version.Info{Major:"1", Minor:"24", GitVersion:"v1.24.3-gke.900", GitCommit:"d5fbbfc5b792af9d522c4b40b3da9595a5512877", GitTreeState:"clean", BuildDate:"2022-07-29T09:22:23Z", GoVersion:"go1.18.3b7", Compiler:"gc", Platform:"linux/amd64"}
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE     VERSION
gke-my-first-cluster-1-default-pool-7ec8fa55-2ldk   Ready    <none>   11m     v1.24.3-gke.900
gke-my-first-cluster-1-default-pool-7ec8fa55-9970   Ready    <none>   7m56s   v1.24.3-gke.900
gke-my-first-cluster-1-default-pool-7ec8fa55-ljrw   Ready    <none>   11m     v1.24.3-gke.900
gke-my-first-cluster-1-default-pool-7ec8fa55-n0hj   Ready    <none>   11m     v1.24.3-gke.900
mr_maulana8@cloudshell:~ (projectz-358108)$
```

Step 2

```
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl create namespace dynatrace
namespace/dynatrace created
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl apply -f https://github.com/Dynatrace/dynatrace-operator/releases/download/v0.8.2/kubernetes.yaml
poddisruptionbudget.policy/dynatrace-webhook created
serviceaccount/dynatrace-activegate created
serviceaccount/dynatrace-kubernetes-monitoring created
serviceaccount/dynatrace-dynakube-oneagent-privileged created
serviceaccount/dynatrace-dynakube-oneagent-unprivileged created
serviceaccount/dynatrace-operator created
serviceaccount/dynatrace-webhook created
customresourcedefinition.apiextensions.k8s.io/dynakubes.dynatrace.com created
clusterrole.rbac.authorization.k8s.io/dynatrace-kubernetes-monitoring created
clusterrole.rbac.authorization.k8s.io/dynatrace-operator created
clusterrole.rbac.authorization.k8s.io/dynatrace-webhook created
clusterrolebinding.rbac.authorization.k8s.io/dynatrace-kubernetes-monitoring created
clusterrolebinding.rbac.authorization.k8s.io/dynatrace-operator created
clusterrolebinding.rbac.authorization.k8s.io/dynatrace-webhook created
role.rbac.authorization.k8s.io/dynatrace-operator created
role.rbac.authorization.k8s.io/dynatrace-webhook created
rolebinding.rbac.authorization.k8s.io/dynatrace-operator created
rolebinding.rbac.authorization.k8s.io/dynatrace-webhook created
service/dynatrace-webhook created
deployment.apps/dynatrace-operator created
deployment.apps/dynatrace-webhook created
mutatingwebhookconfiguration.admissionregistration.k8s.io/dynatrace-webhook created
validatingwebhookconfiguration.admissionregistration.k8s.io/dynatrace-webhook created
mr_maulana8@cloudshell:~ (projectz-358108)$
```

Execute every command that you have copied before one by one and make sure there is no error on every step. If there is an error in previous step, the next step will be fail.

1. kubectl create namespace Dynatrace

2. kubectl apply -f https://github.com/Dynatrace/dynatrace-operator/releases/download/v0.7.2/kubernetes.yaml

Step 3

```
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl -n dynatrace wait pod --for=condition=ready --selector=app.kubernetes.io/name=dynatrace-operator,app.kubernetes.io/component=webhook --timeout=3
00s
pod/dynatrace-webhook-fdf9d6c8-hfwtf condition met
pod/dynatrace-webhook-fdf9d6c8-thvbp condition met
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl apply -f dynakube.yaml
secret/gke-petclinic created
Warning: annotation 'feature.dynatrace.com/enable-activegate-authtoken' is deprecated, use 'feature.dynatrace.com/activegate-authtoken' instead
dynakube.dynatrace.com/gke-petclinic created
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl get pod -n dynatrace
NAME                                  READY   STATUS              RESTARTS   AGE
dynatrace-operator-85ccf4c84f-5w772   1/1     Running             0          4m5s
dynatrace-webhook-fdf9d6c8-hfwtf      1/1     Running             0          4m4s
dynatrace-webhook-fdf9d6c8-thvbp      1/1     Running             0          4m4s
gke-petclinic-activegate-0            0/1     Pending             0          16s
gke-petclinic-oneagent-7n6nh          0/1     ContainerCreating   0          17s
gke-petclinic-oneagent-sv5tb          0/1     ContainerCreating   0          17s
gke-petclinic-oneagent-vn5bx          0/1     ContainerCreating   0          17s
gke-petclinic-oneagent-w2j45          0/1     ContainerCreating   0          17s
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl get pod -n dynatrace
NAME                                  READY   STATUS    RESTARTS   AGE
dynatrace-operator-85ccf4c84f-5w772   1/1     Running   0          6m25s
dynatrace-webhook-fdf9d6c8-hfwtf      1/1     Running   0          6m24s
dynatrace-webhook-fdf9d6c8-thvbp      1/1     Running   0          6m24s
gke-petclinic-activegate-0            0/1     Pending   0          2m36s
gke-petclinic-oneagent-7n6nh          1/1     Running   0          2m37s
gke-petclinic-oneagent-sv5tb          1/1     Running   0          2m37s
gke-petclinic-oneagent-vn5bx          1/1     Running   0          2m37s
gke-petclinic-oneagent-w2j45          1/1     Running   0          2m37s
mr_maulana8@cloudshell:~ (projectz-358108)$
```

3. kubectl -n dynatrace wait pod --for=condition=ready --selector=app.kubernetes.io/name=dynatrace-operator,app.kubernetes.io/component=webhook --timeout=300s

4. kubectl apply -f dynakube.yaml

Wait for Dynatrace Operator components to finish initialization.

The downloaded YAML file is a basic version of the DynaKube custom resource definition. To adapt values to your specific needs, please check Referrence

If you already have dedicated Activegate, then you need to remove activegate line from dynakube.yaml

Step 4

Dynatrace will discover all Kubernetes Nodes as a Host.

Hosts

Hosts

Pin to dashboard ···

Filtered by Tag: [GoogleCloud]gke-my-fi... ✕

Quick filters

All filter options available in the filter field

Problem impact

- (•) Any
- ( ) Impacted
- ( ) Not impacted

State

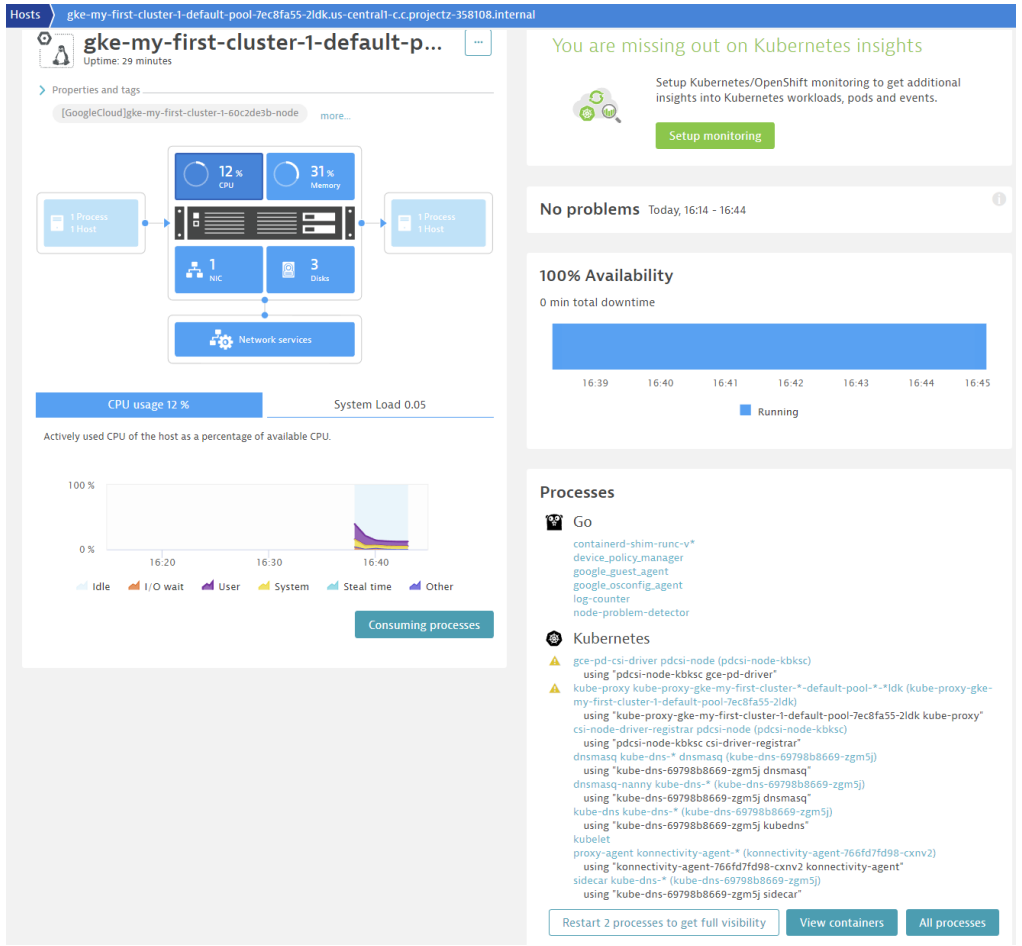
- (•) Any
- ( ) Offline
- ( ) Running

4 Hosts

Name	Operating system	Virtualization	CPU usage	Memory usage	Disk latency	Network traffic
gke-my-first-cluster-1-default-pool-7ec8fa55-2ldk.us-central1-c.c.projectz-358108.internal	Linux	Google Compute Engine	12 %	31 % of 1.64 GiB	8.44 ms	65.4 kbit/s
gke-my-first-cluster-1-default-pool-7ec8fa55-9970.us-central1-c.c.projectz-358108.internal	Linux	Google Compute Engine	12 %	32 % of 1.64 GiB	32 ms	96.6 kbit/s
gke-my-first-cluster-1-default-pool-7ec8fa55-ljrw.us-central1-c.c.projectz-358108.internal	Linux	Google Compute Engine	12 %	33 % of 1.64 GiB	6.56 ms	81.8 kbit/s
gke-my-first-cluster-1-default-pool-7ec8fa55-n0hj.us-central1-c.c.projectz-358108.internal	Linux	Google Compute Engine	14 %	33 % of 1.64 GiB	10.2 ms	127 kbit/s

Step 5

Some process (pods) may need to restart in order to enable Deep Monitoring feature



Monitor Kubernetes API Cluster

with Activegate VM

Step 1

1. Create a **kubernetes-monitoring-service-account.yaml** file with the following content.

```yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: dynatrace-monitoring
  namespace: dynatrace
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: dynatrace-monitoring-cluster
rules:
- apiGroups:
  - ""
  - batch
  - apps
  - apps.openshift.io
  resources:
  - nodes
  - pods
  - namespaces
  - deployments
  - replicasets
  - deploymentconfigs
  - replicationcontrollers
  - jobs
  - cronjobs
  - statefulsets
  - daemonsets
  - events
  - resourcequotas
  - pods/proxy
  - services
  verbs:
  - list
  - watch
  - get
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: dynatrace-monitoring-cluster
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: dynatrace-monitoring-cluster
subjects:
- kind: ServiceAccount
  name: dynatrace-monitoring
  namespace: dynatrace
```

Step 2

2. Apply the file.

```
kubectl apply -f kubernetes-monitoring-service-account.yaml
```

3. Get the Kubernetes API URL.

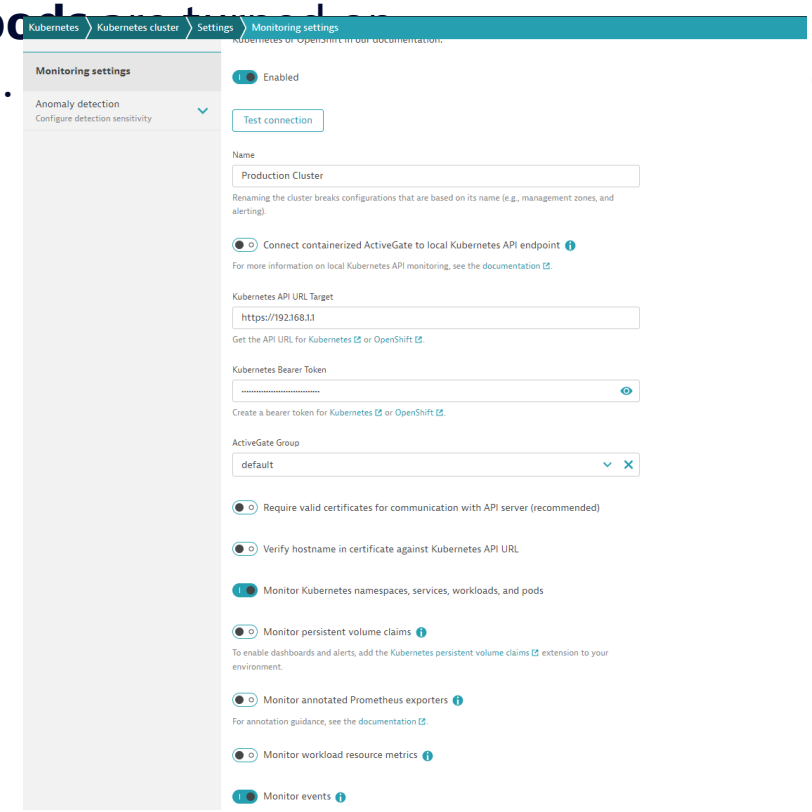
```
kubectl config view --minify -o jsonpath='{.clusters[0].cluster.server}'
```

4. Get the bearer token.
kubectl get secret $(kubectl get sa dynatrace-monitoring -o jsonpath='{.secrets[0].name}' -n dynatrace) -o jsonpath='{.data.token}' -n dynatrace | base64 –decode

5. Copy the output from command no 3 and 4 because it will be used for another step

Step 3

6. In the Dynatrace menu, go to **Kubernetes** -> **Connect manually**
7. Provide a **Name**, the **Kubernetes API URL target**, and the **Kubernetes bearer token** for the Kubernetes cluster (Paste from previous command output).
8. Make sure **Monitor events** and **Monitor Kubernetes namespaces, services, workloads, and pods** are turned on
9. **Changes**



Result

You will be able to see your cluster utilization in less than 5 minutes. It will take sometimes to collect data from Kubernetes API Cluster.

Kubernetes

Kubernetes clusters

Overview of Kubernetes clusters

Dashboards | View all workloads | Connect manually | Connect automatically via Dynatrace Operator

You are missing out on monitoring data for 3 of your configured Kubernetes clusters. Please check the settings of the corresponding clusters. Disable the toggle below to show all clusters.

Filter by

Only show clusters with data in the selected timeframe (Sep 21 16:28 - 16:58)

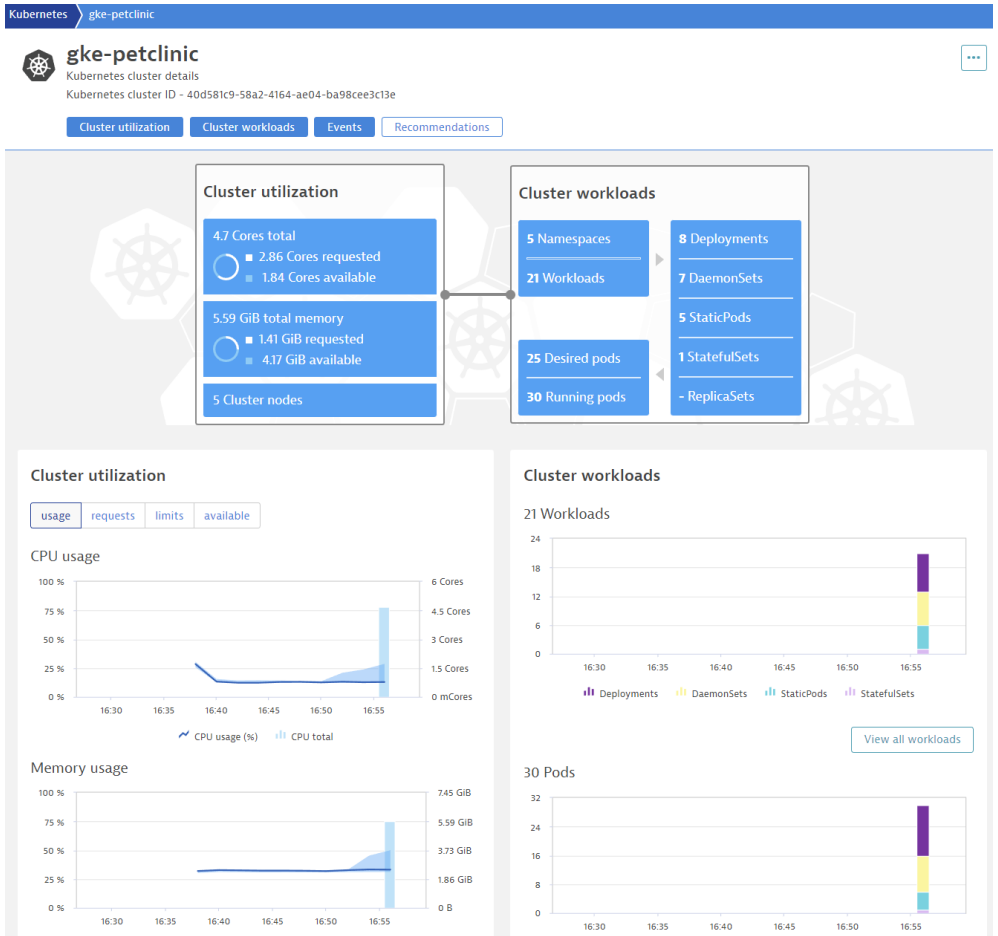
1 Cluster

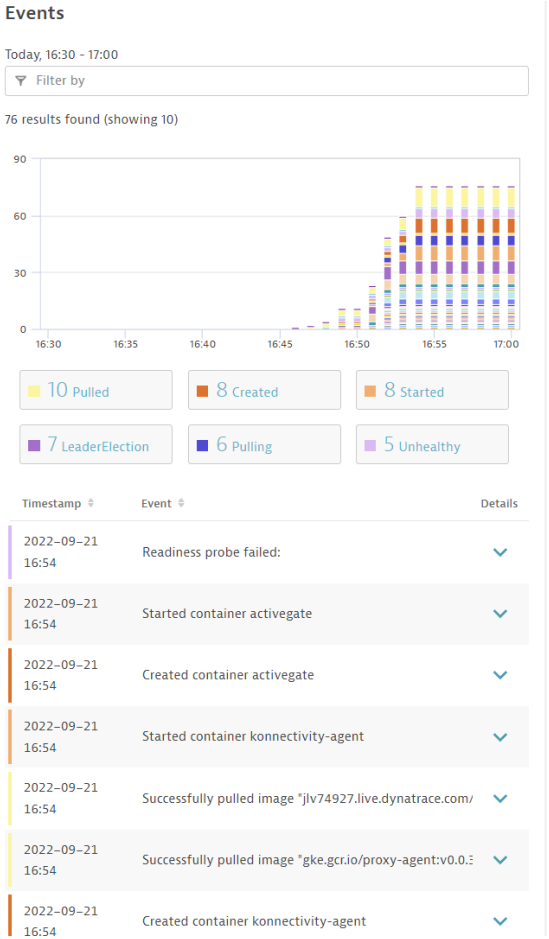
Name	Namespaces	Workloads	Nodes	CPU total	Memory requested	Memory total	Actions
gke-petclinic	5 Namespaces	21 Workloads	5 Nodes	4.7 Cores	1.41 GiB	5.59 GiB	•••

1

Result

Kubenetes Cluster events also can monitored with Dynatrace as long as it is enabled





Uninstall Dynatrace Operator

Uninstall Dynatrace Operator

To uninstall Dynatrace Operator, simple run below command and wait until all component deleted.

```
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl delete -f https://github.com/Dynatrace/dynatrace-operator/releases/download/v0.8.2/kubernetes.yaml
poddisruptionbudget.policy "dynatrace-webhook" deleted
serviceaccount "dynatrace-activegate" deleted
serviceaccount "dynatrace-kubernetes-monitoring" deleted
serviceaccount "dynatrace-dynakube-oneagent-privileged" deleted
serviceaccount "dynatrace-dynakube-oneagent-unprivileged" deleted
serviceaccount "dynatrace-operator" deleted
serviceaccount "dynatrace-webhook" deleted
customresourcedefinition.apiextensions.k8s.io "dynakubes.dynatrace.com" deleted
clusterrole.rbac.authorization.k8s.io "dynatrace-kubernetes-monitoring" deleted
clusterrole.rbac.authorization.k8s.io "dynatrace-operator" deleted
clusterrole.rbac.authorization.k8s.io "dynatrace-webhook" deleted
clusterrolebinding.rbac.authorization.k8s.io "dynatrace-kubernetes-monitoring" deleted
clusterrolebinding.rbac.authorization.k8s.io "dynatrace-operator" deleted
clusterrolebinding.rbac.authorization.k8s.io "dynatrace-webhook" deleted
role.rbac.authorization.k8s.io "dynatrace-operator" deleted
role.rbac.authorization.k8s.io "dynatrace-webhook" deleted
rolebinding.rbac.authorization.k8s.io "dynatrace-operator" deleted
rolebinding.rbac.authorization.k8s.io "dynatrace-webhook" deleted
service "dynatrace-webhook" deleted
deployment.apps "dynatrace-operator" deleted
deployment.apps "dynatrace-webhook" deleted
mutatingwebhookconfiguration.admissionregistration.k8s.io "dynatrace-webhook" deleted
validatingwebhookconfiguration.admissionregistration.k8s.io "dynatrace-webhook" deleted
mr_maulana8@cloudshell:~ (projectz-358108)$
```

Make sure all the Dynatrace namespace or resource all delete

```
mr_maulana8@cloudshell:~ (projectz-358108)$ kubectl get pod -n dynatrace
No resources found in dynatrace namespace.
mr_maulana8@cloudshell:~ (projectz-358108)$
```

dynatrace

CLOUD DONE RIGHT